

# Real-Time Student Mental Wellness Predictor Using Multi-Source Data and Explainable Machine Learning

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## **Abstract:**

Mental health disorders, particularly anxiety and depression, have emerged as a critical global crisis among the student population, with prevalence rates escalating sharply in the post-pandemic era. According to the World Health Organization, approximately one in four university students worldwide experiences a diagnosable mental health condition, yet the majority remain unidentified until the condition reaches a severe and treatment-resistant stage. Traditional screening mechanisms—predominantly periodic self-report questionnaires such as PHQ-9 and GAD-7—are constrained by infrequent administration, recall bias, social desirability effects, and an inability to detect gradual psychological deterioration between assessment windows. These limitations underline the urgent need for intelligent, continuous, and interpretable monitoring systems capable of providing early-stage risk identification. This paper proposes MindGuard, a Real-Time Student Mental Wellness Prediction System that addresses these shortcomings through a three-pronged technical approach. First, a Hybrid Stacking Ensemble architecture is developed, combining two complementary base learners—Logistic Regression and XGBoost—under a Logistic Regression meta-learner that adaptively learns optimal combination weights from out-of-fold cross-validated predictions. Logistic Regression as a base learner provides a calibrated linear probabilistic boundary, while XGBoost captures complex non-linear feature interactions through gradient-boosted trees; together their predictions form a rich stacked feature representation that significantly improves generalization over either model trained independently. Second, to overcome the interpretability deficit inherent in black-box ensemble models, two complementary Explainable AI (XAI) techniques are integrated: SHAP (SHapley Additive exPlanations) for global feature importance grounded in cooperative game theory, and LIME (Local Interpretable Model-agnostic Explanations) for locally faithful per-instance explanations. Together, these provide counselors and administrators with transparent, actionable insights into each prediction. Third, the complete pipeline is deployed as a Flask RESTful web application that collects student data via a structured digital questionnaire, processes it through the ensemble model, and delivers real-time risk-level classification into four clinically aligned tiers: Low, Moderate, High, and Critical Risk. Experiments conducted on a composite dataset of 1,387 student records—validated by licensed clinical psychologists using PHQ-9 and GAD-7 thresholds—demonstrate that the proposed system achieves 93.8% classification accuracy, a macro F1-score of 0.925, and an MCC of 0.912, outperforming all individual baseline models and a voting ensemble alternative with statistical significance ( $p < 0.05$ ). SHAP analysis identifies sleep quality and academic stress as the dominant predictors of mental wellness risk, consistent with established clinical literature. The system achieves an average inference latency of 1.2 seconds per prediction, confirming real-time suitability. These results underscore the potential of MindGuard as a proactive, transparent, and scalable mental health intervention tool for educational institutions.

## **Keywords:**

*Mental Health Prediction; Hybrid Stacking Ensemble; XGBoost; SHAP; LIME; Explainable AI; Flask; Student Wellness; Depression Detection; Anxiety Classification.*